Fiscal Year:	FY 2023	**Task Last Updated:**	FY 12/02/2022
PI Name:	Shelhamer, Mark Sc.D.		
Project Title:	Assessment of Otolith Function and Asymmetry as a Corollary to Critical Sensorimotor Performance in Missions of Various Durations		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Baltimore	**State:**	MD
Zip Code:	21205-1832	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	02/07/2019	**End Date:**	02/07/2033
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/7/2033 per L. Barnes-Moten/JSC and NSSC information (Ed., 5/18/21) NOTE: End date changed to 6/6/2021 per NSSC information (Ed., 11/9/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schubert, Michael Ph.D. (Johns Hopkins University)		
Grant/Contract No.:	80NSSC19K0487		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose a simple and rapid measure of otolith asymmetry -- a basic low-level neurovestibular function that is altered in spaceflight. Our assessment apparatus and procedure was developed with previous Human Research Program (HRP) support, and has undergone validation testing and is currently in use in an unrelated Department of Defense (DoD) study. It provides a measure of binocular alignment (vertical and torsional), which is a reflection of underlying otolith asymmetry. While this measure might be important in itself (since it reflects a form of central neural compensation that might be challenged in spaceflight), it could also provide a simple alternative to locomotion and posture testing if it is found to correlate with those measures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The technology and procedures for this project are being used in DoD projects for the assessment of vestibular function in military personnel. The technology has potential clinical applications.
Task Progress:	This project was absorbed into the CIPHER (Complement of Integrated Protocols for Human Research) Vestibular Health IRB (Institutional Review Board) project as part of a broader re-scoping of CIPHER early in 2022. Online integrated consent briefings were supported, and a presentation was made at the online HRP (Human Research Program) Investigators Workshop in February 2022. In the fall of 2021, we started a set of experiments in the lab to provide further baseline data by which to evaluate the upcoming flight data from this project. In these trials, subjects are evaluated for ocular alignment (OA) while standing on a hard floor, standing on foam, while seated, and while making head movements. This will help us evaluate the impact of postural demands on ocular alignment; we conjecture that the heightened demand while standing will call for a more precise compensation for any latent otolith asymmetry, causing a reduction in misalignment in that condition compared to seated. Eight subjects have been tested to date on the entire protocol (12-15 planned in total), and analysis is ongoing. Initial observations show that performance is variable across subjects (as expected). Subjects unsurprisingly have the most difficulty when making pitching head movements. Otherwise, there are few notable differences between conditions. A recent change was made to the OA software. Now, we collect information on the time that it takes a subject to perform each trial as a way to assess the level of confidence, motor control, and related factors. A portion of the protocol for this project was modified and extended for use as an educational demonstration at Space Center Houston. This enables the collection of a large set of normative data (N=157 currently) with which to refine our analyses and which can provide context for the interpretation of results from astronauts (commercial and professional). This project also includes a component dedicated to data-integration of the other CIPHER projects. While awaiting CIPHER flight data, we have been performing several preparatory activities so that we can be better prepared to organize and analyze that data: 1. Careful examination of the CIPHER complement to understand the measures and data that will be obtained, their modalities and form (numeric, imaging, etc.), and their sampling schedules. This will lead to an initial “binning” of the data: identification of those measures that can be considered to have been acquired at the same time point, taking into account the time course of change of the various physiological systems under study. This will then lead to an initial set of hypotheses and a statistical plan. 2. Identification of related datasets that are already available that could help address some of the same questions as CIPHER (i.e., are from the same or similar body systems), and would permit a longitudinal (personalized) investigation. The latter point is crucial since a key point of CIPHER integration is that data can be attributed to each individual astronaut test subject, allowing personalized cross-disciplinary study. Potential datasets under consideration include those from Mars 500, published data from the early space program up through Skylab (when data were published in identifiable/attributable form), and LSDA (Life Sciences Data Archive) data from STS (Space Transportation System) missions (under consultation with LSDA leadership). 3. Text analytics on the CIPHER proposals to identify common themes, methods, or approaches across studies. First, we examined reference lists in each CIPHER proposal for citations to common authors across proposals that might be unexpected (if proposals are in different disciplines) and might indicate unforeseen scientific connections. Second, networks were created to visualize how proposals are linked to each other through common terms relevant to HRP, such as its set of established hazards, risks, and environmental metadata. Third, Latent Dirichlet Allocation (LDA) was applied to identify underlying concepts that might connect different proposals. This investigation to date has provided some insights and few surprises but establishes an infrastructure by which to understand and interpret CIPHER within the larger set of space life-sciences concerns.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Macaulay TR, Wood SJ, Bollinger A, Schubert MC, Shelhamer M, Bishop MO, Reschke MF, Clément G. "Comparison of asymmetry between perceptual, ocular, and postural vestibular screening tests." Brain Sci. 2023 Jan 23;13(2):189. https://doi.org/10.3390/brainsci13020189 ; PMID: 36831732; PMCID: PMC9954277 , Jan-2023